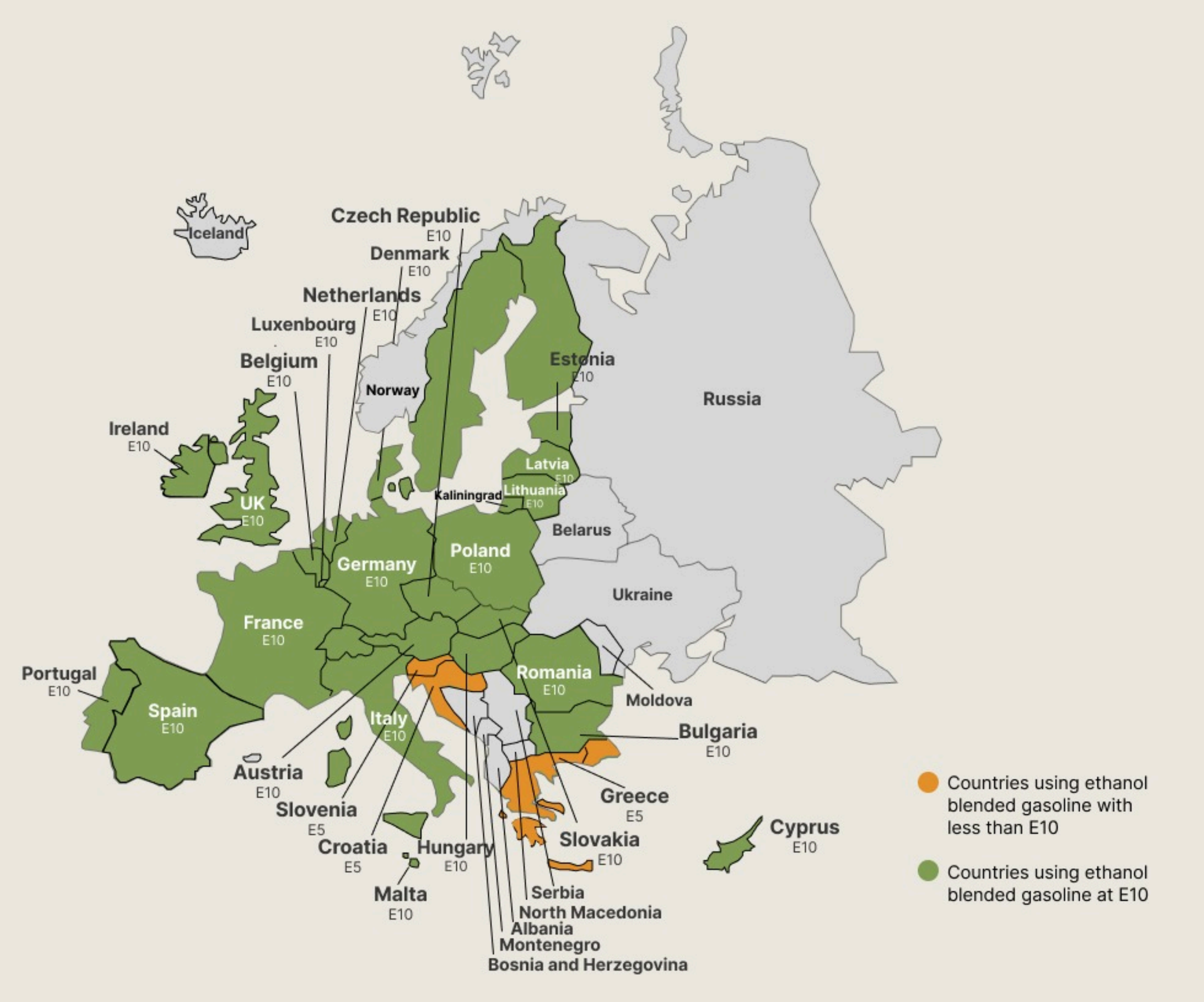
Ethanol Blending in Gasoline - Ireland

September, 2025

Ethanol Blending in Europe

There are important fuel quality and environmental impact of vehicle emission challanges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

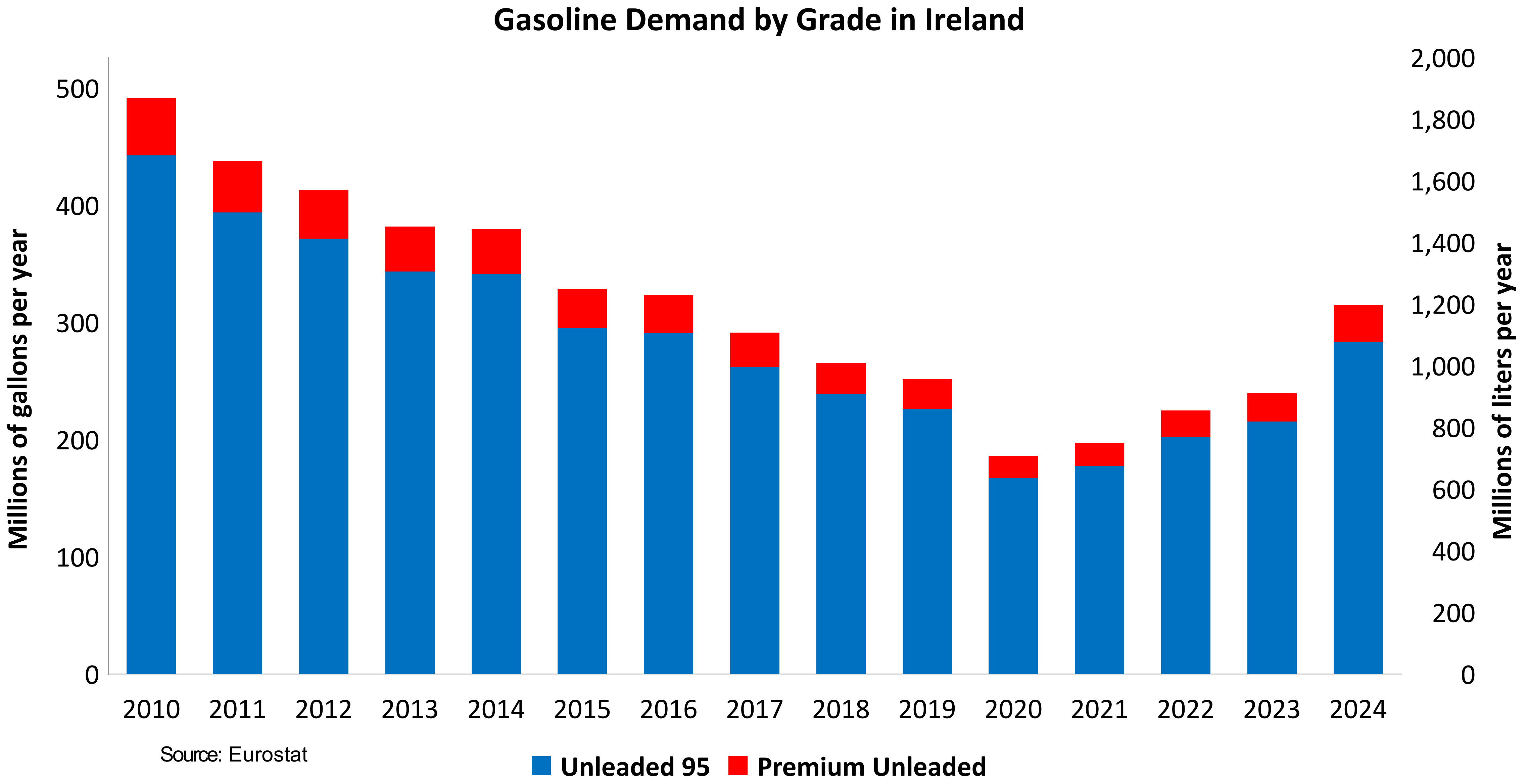
Ethanol Gasoline Blending in Ireland

In 2024, gasoline consumption in Ireland was 1,197 million liters (316 million gallons) and growth rate was 4.6%. Gasoline production and net imports were 676 and 521 million liters (179 and 138 million gallons) correspondingly. Ireland complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.3 RON.

In 2024, ethanol consumption in Ireland was 60 million liters (16 million gallons) representing 5.0% of gasoline demand. Ethanol production and Net Imports were 22 and 39 million liters (16 and 10 million gallons) correspondingly. Ethanol source is Cereals 100%.

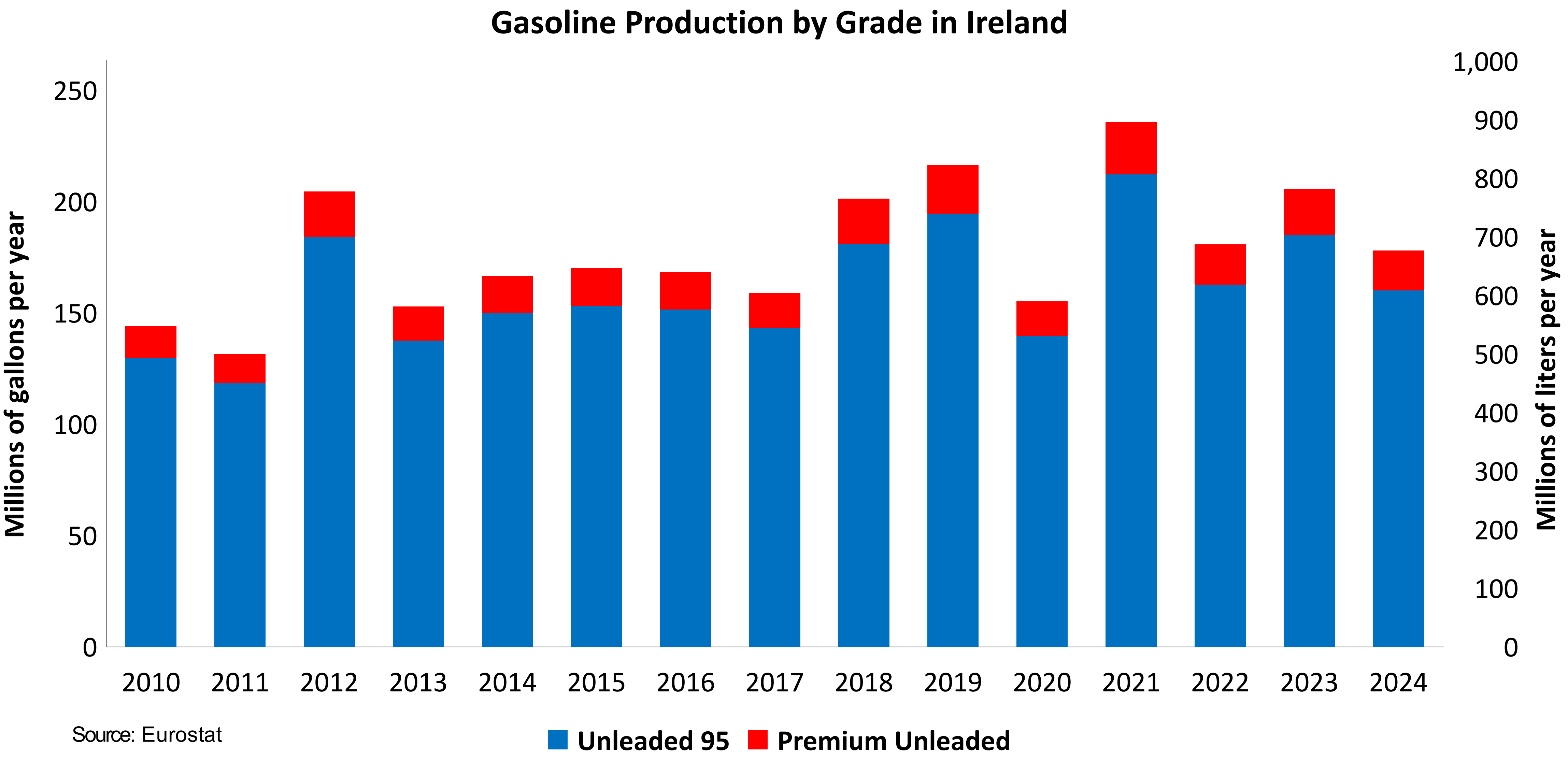
Source: Eurostat, European Commision

Gasoline Demand in Ireland



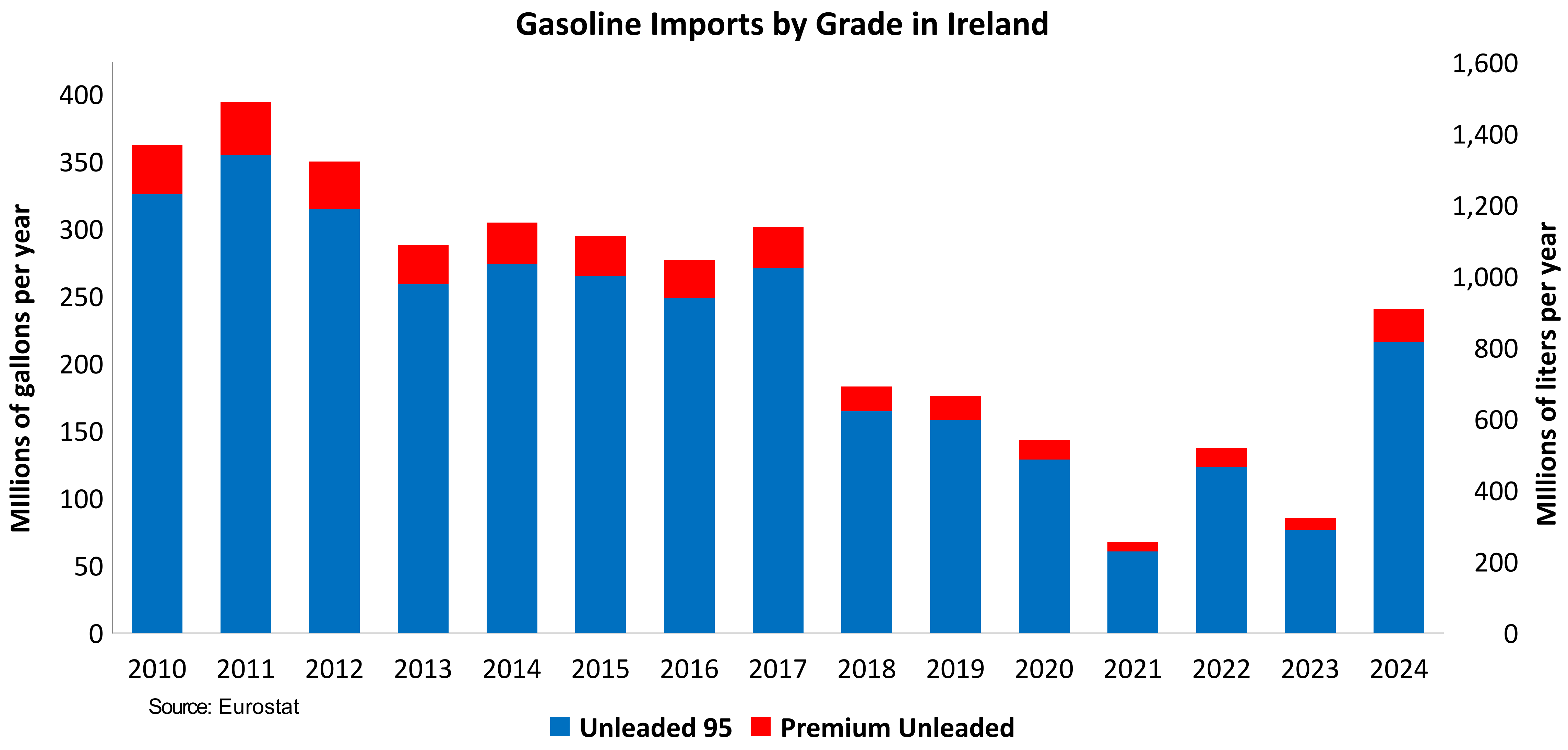
Source: Eurostat

Gasoline Production in Ireland

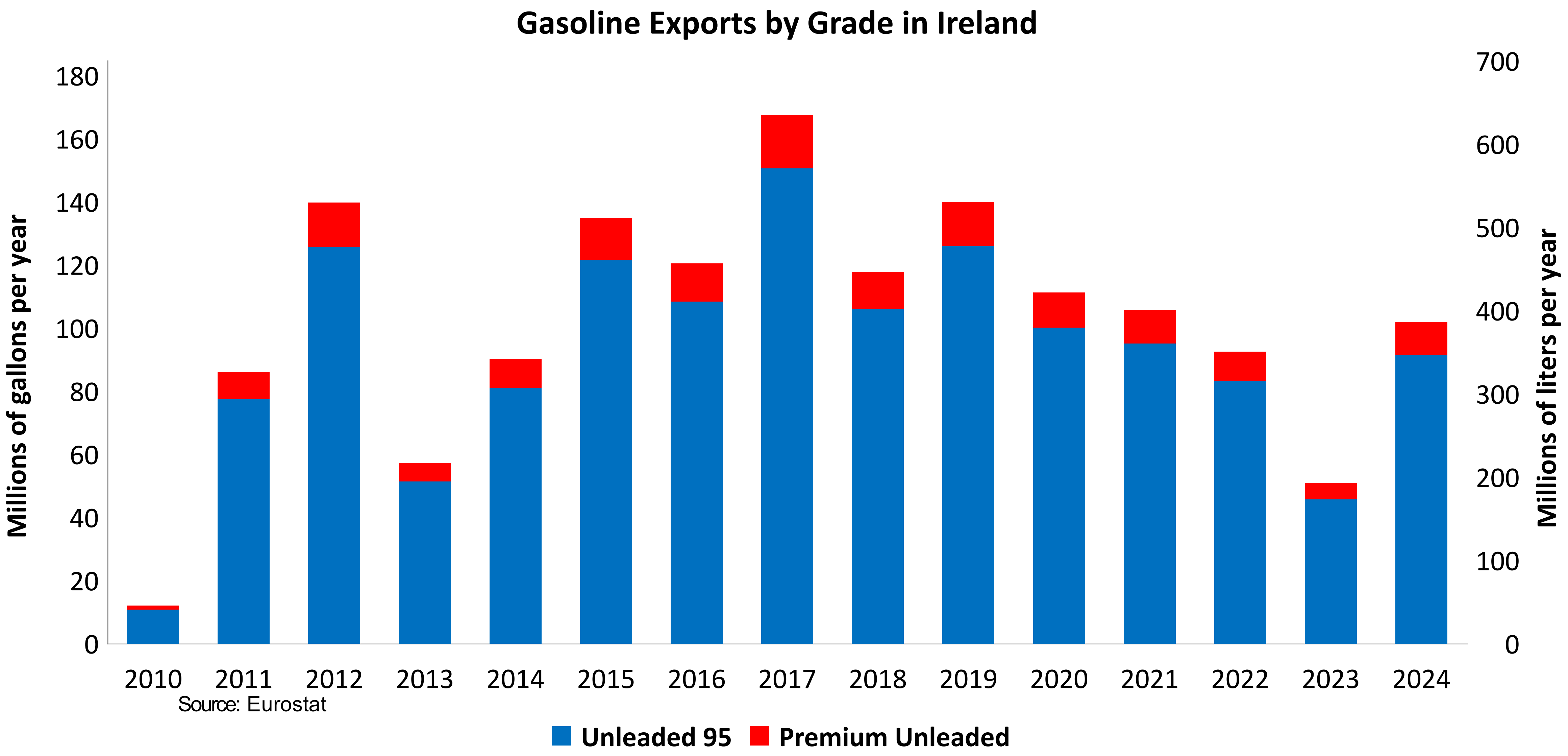


Source: Eurostat

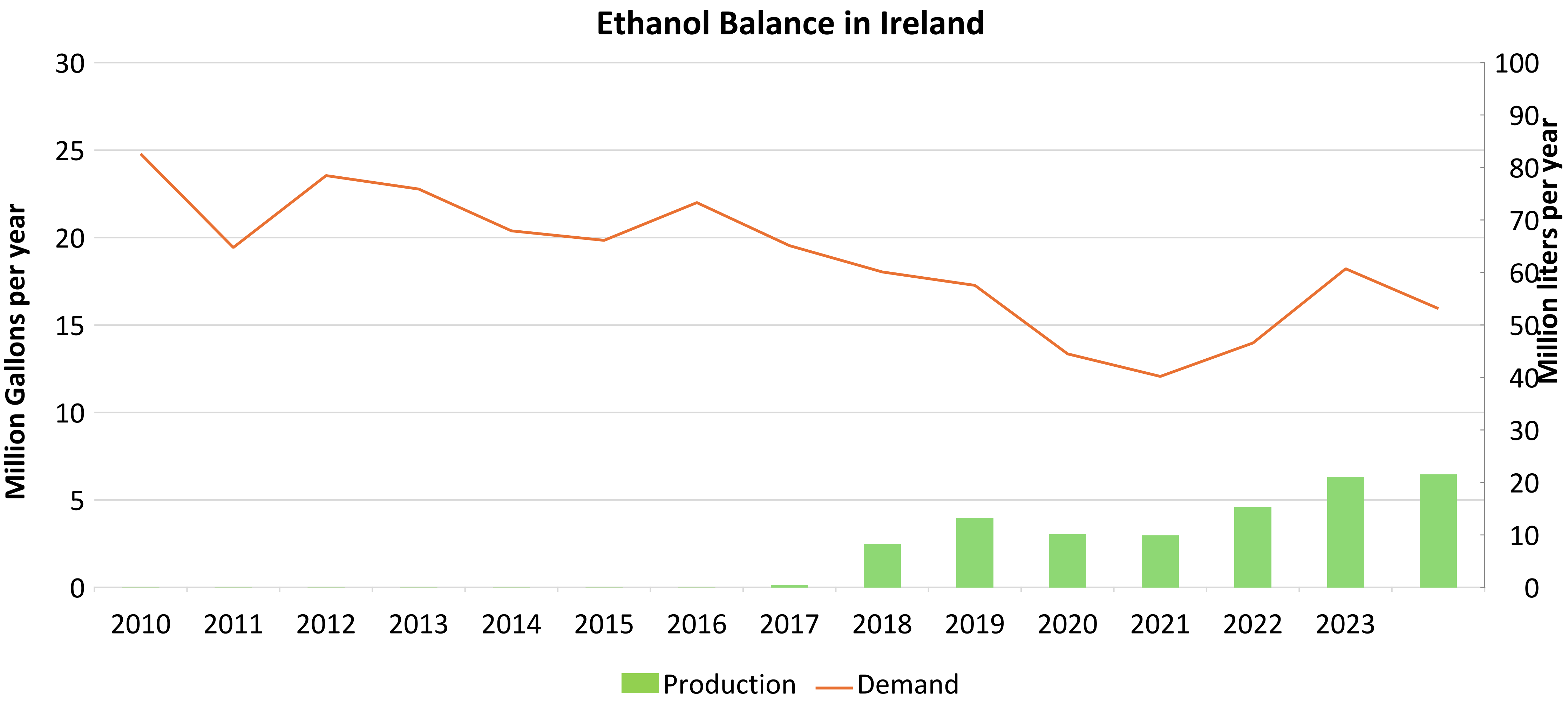
Gasoline Imports in Ireland



Gasoline Exports in Ireland



Ethanol Balance in Ireland



Source: Eurostat

Gasoline Quality in Ireland

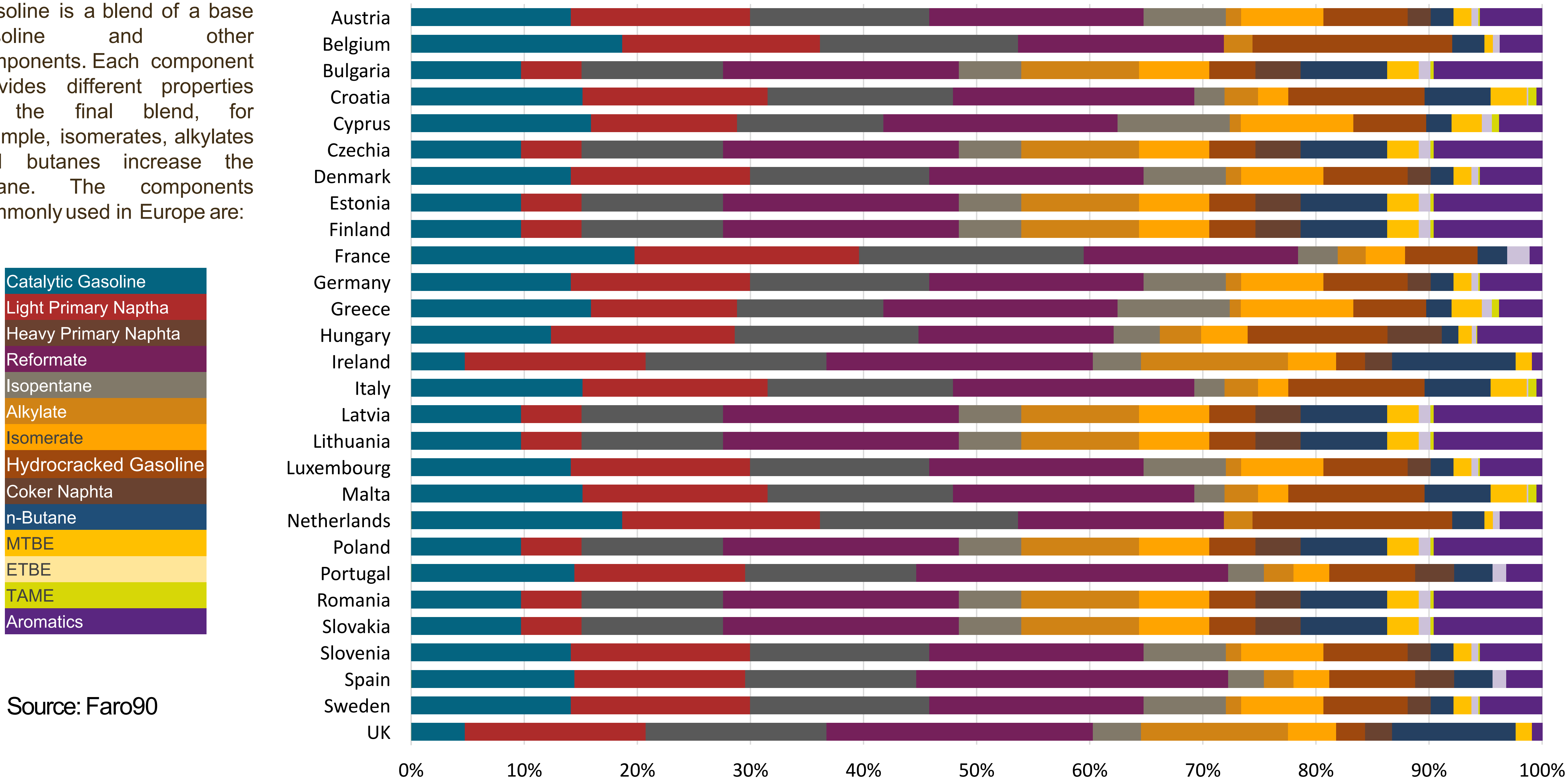
Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	21.0							
Advanced Biofuels (%cal)	1.0							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Gasoline Component Blending in Europe

Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Europe are:



Source: Faro90

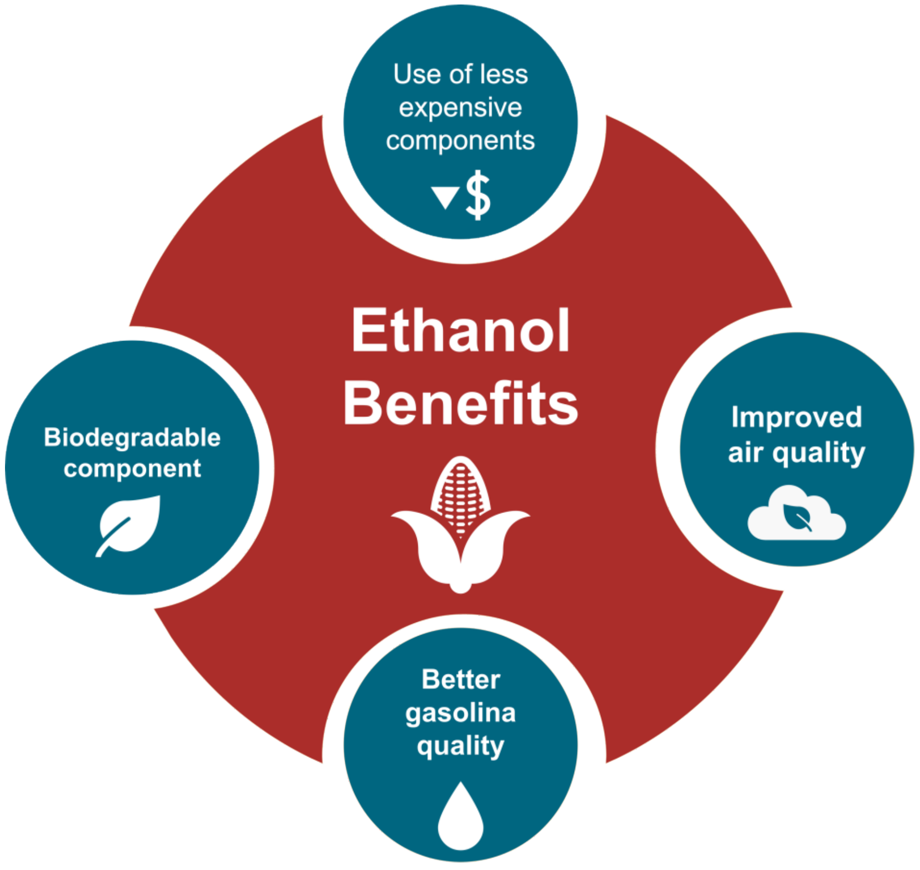
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component.The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives.To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

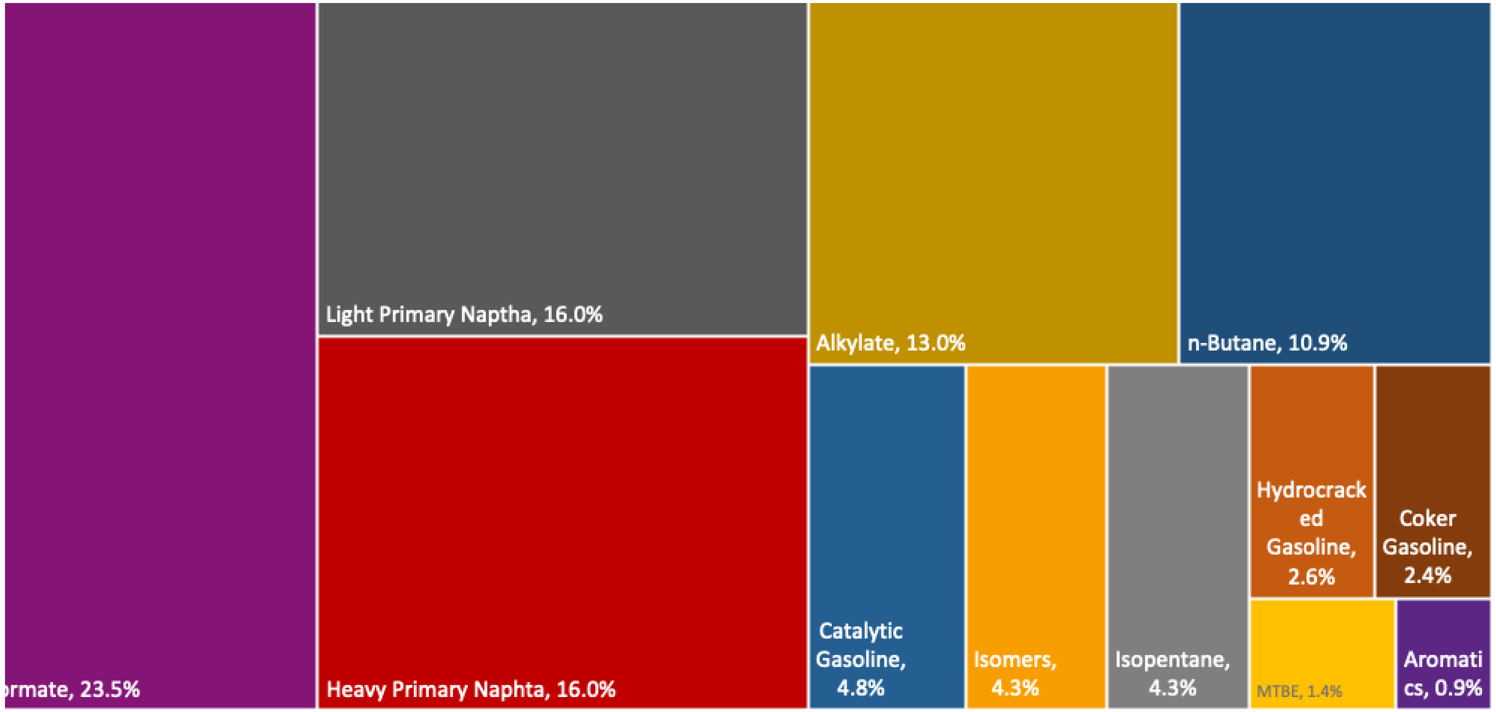
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

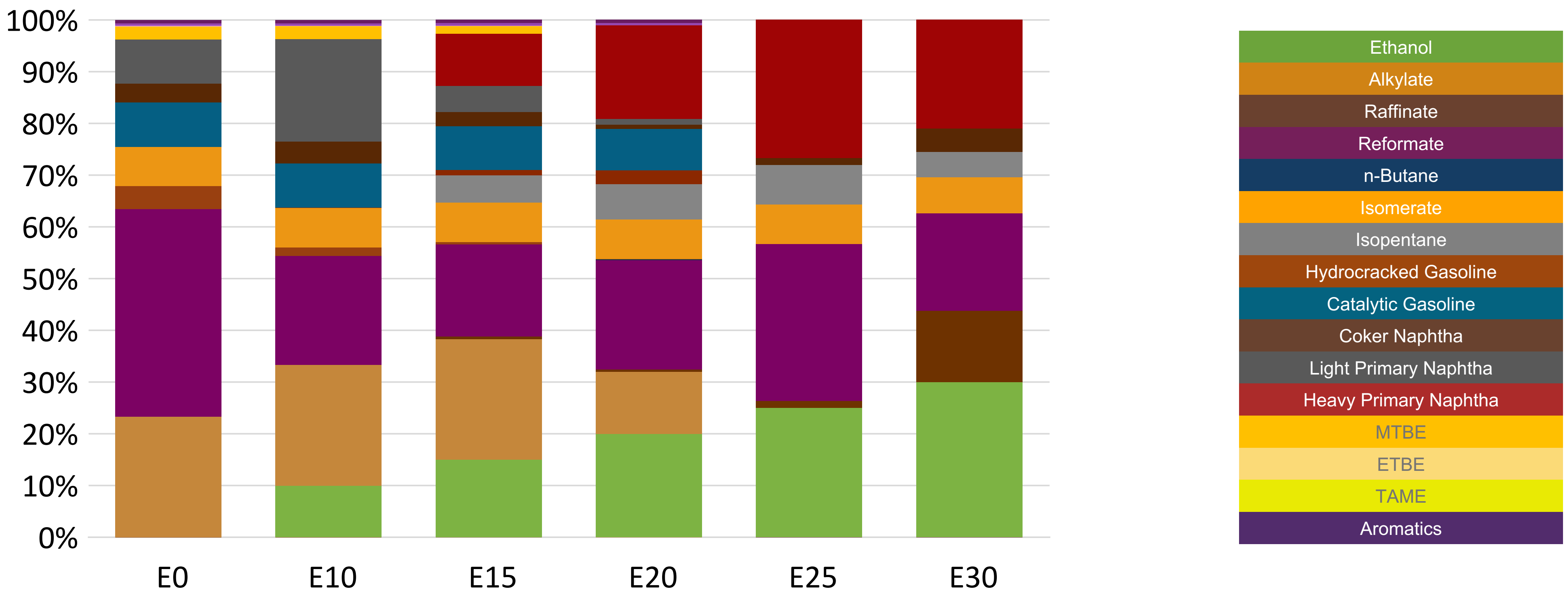


Ireland Available Blending Components



Source: Faro90

Ireland – Gasoline 95 – Constant Octane

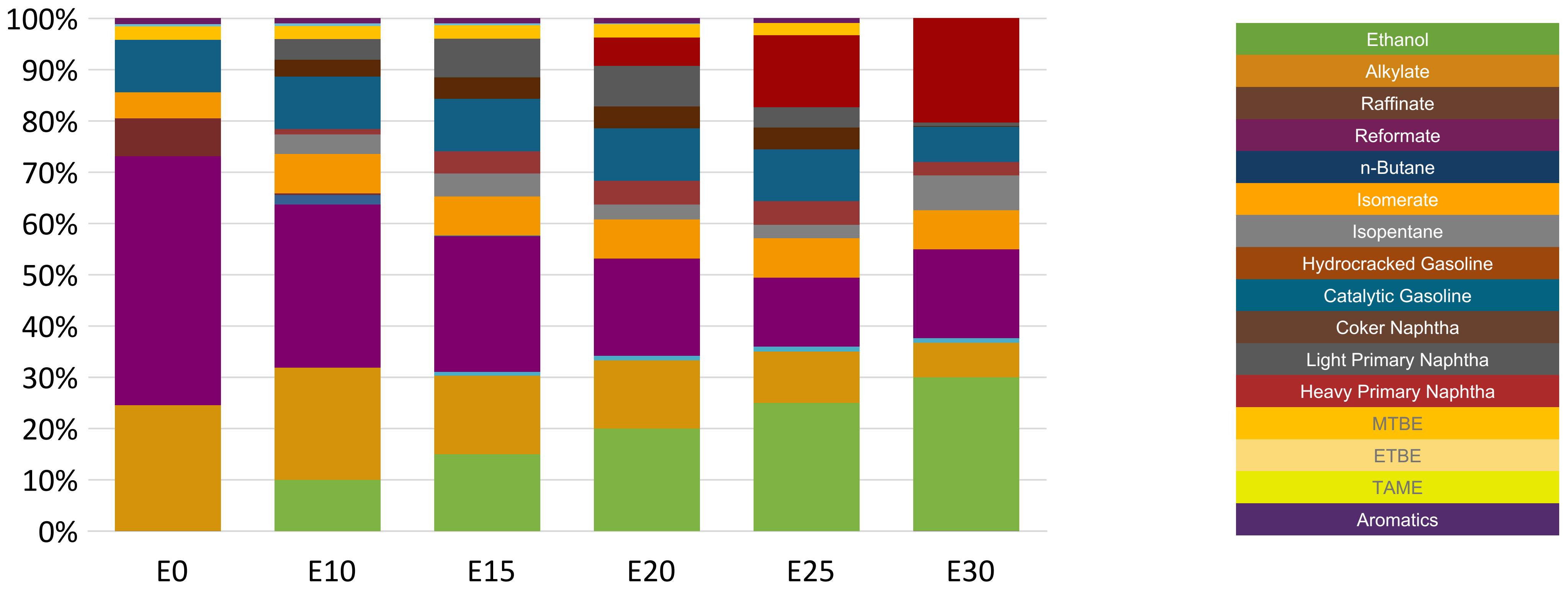


Octane (RON)	95.0	95.0	95.0	95.0	95.4	95.8
Price (US$/gal)	$ 2.317	$ 2.187	$ 2.105	$ 1.996	$ 1.886	$ 1.821

Average CIF 2024 Prices

Source: Faro90

Ireland - Gasoline 98 - Constant Octane

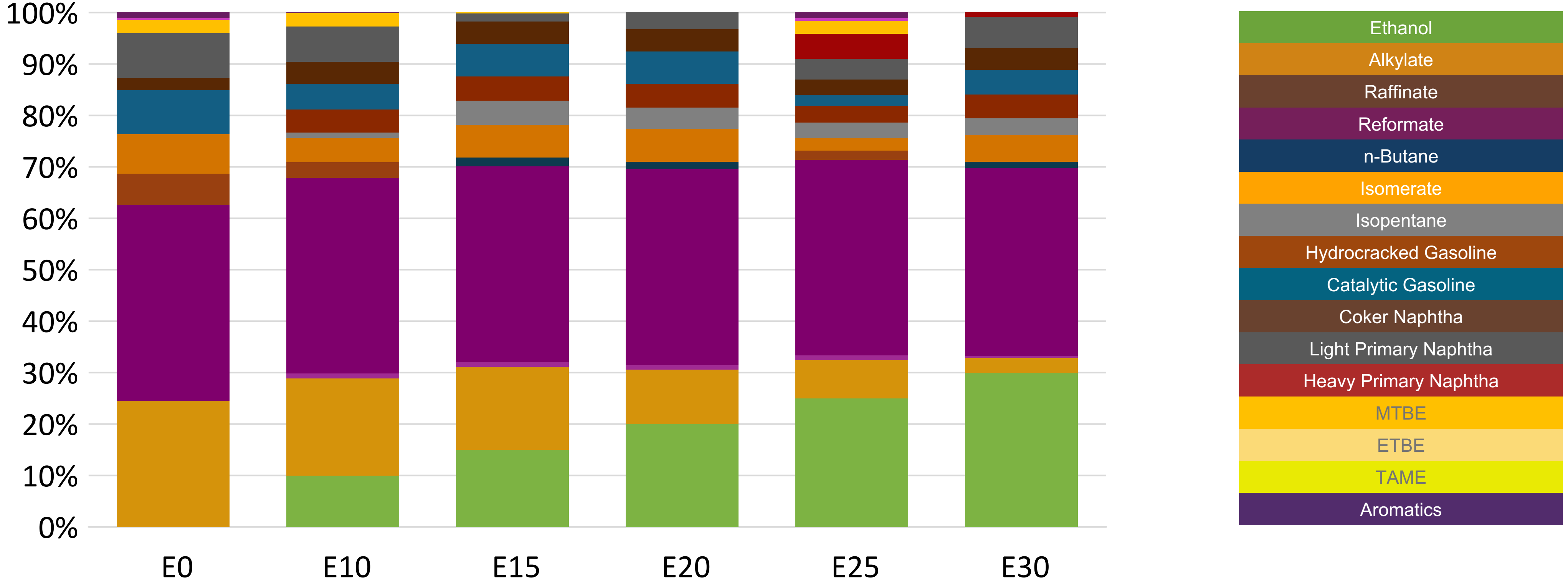


Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US$/gal)	$ 2.440	$ 2.317	$ 2.240	$ 2.157	$ 2.062	$ 1.967

Average CIF 2024 Prices

Source: Faro90

Ireland – Gasoline 95 – Increased Octane

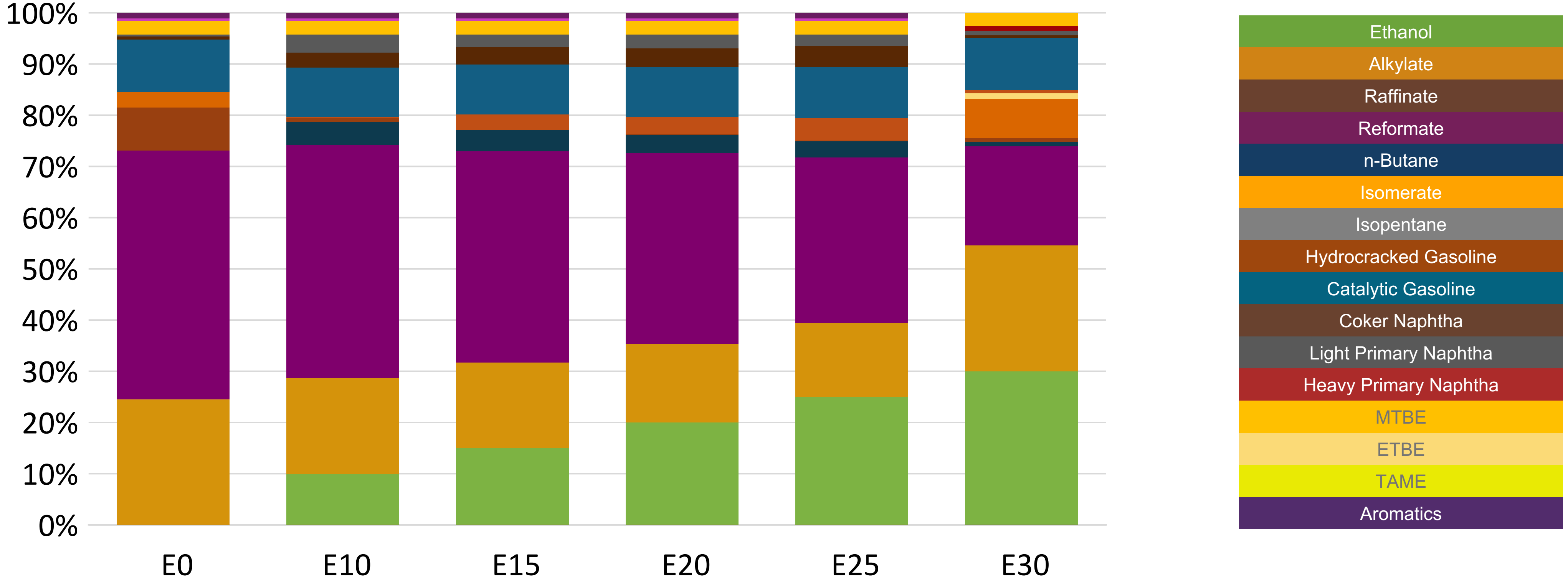


Octane (RON)	95.0	97.9	99.3	100.9	102.6	103.9
Price (US$/gal)	$ 2.301	$ 2.301	$ 2.301	$ 2.301	$ 2.301	$ 2.301

Average CIF 2024 Prices

Source: Faro90

Ireland – Gasoline 98 – Increased Octane



Octane (RON)	98.0	100.7	102.0	103.3	104.5	105.0
Price (US$/gal)	$ 2.425	$ 2.425	$ 2.425	$ 2.425	$ 2.425	$ 2.425

Average CIF 2024 Prices

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

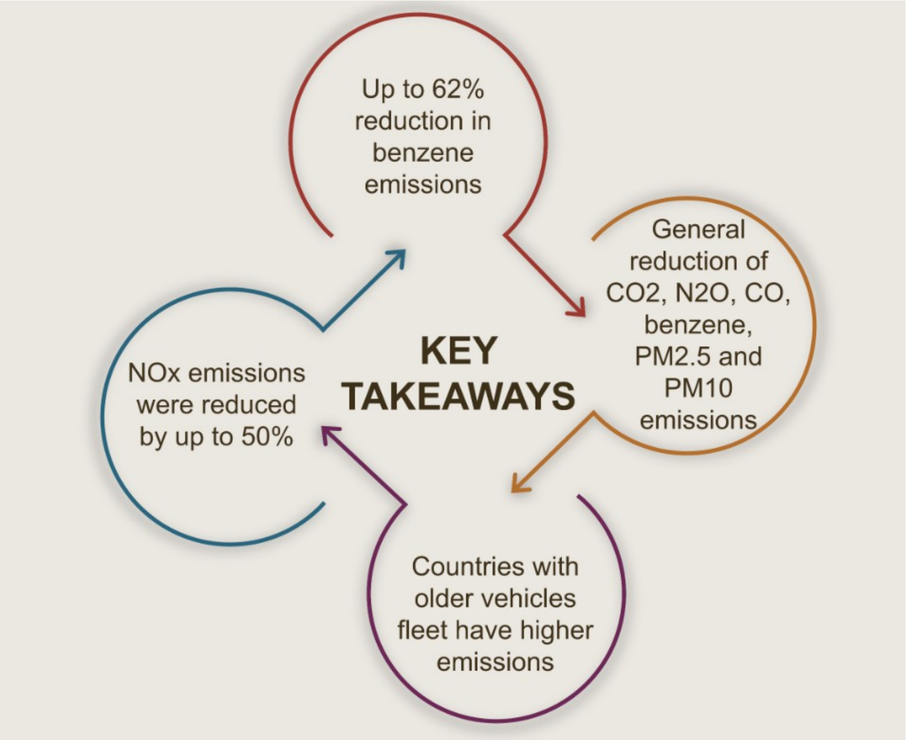
The model used in this analysis takes as a reference the **International Vehicle Emissions Model (IVE).**
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Main Results



*Source: Bureau of transportation statistics.

Gasoline Vehicle Fleet - Ireland

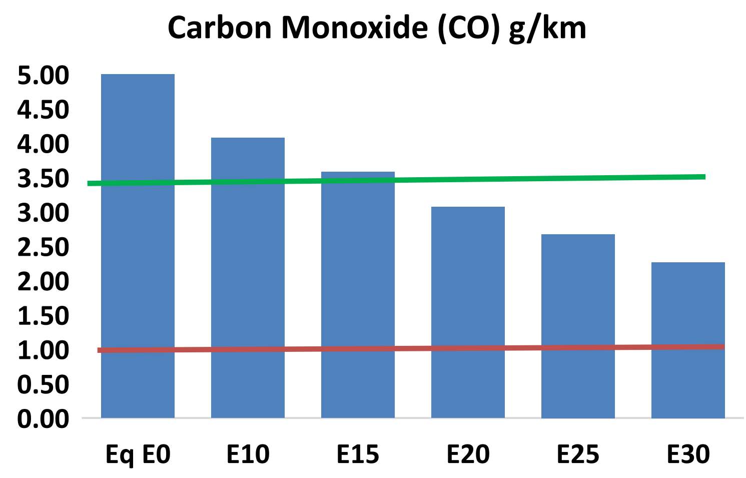
Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	665
		Single FI	3-Ways		>8 years	4,144
		Multi FI	3-Ways		>8 years	182,350
			Euro III		>8 years	74,376
			Euro IV		>8 years	126,900
			Euro V		>8 years	45,876
			Euro 6		>8 years	110,643
					4-8 years	73,652
					<4 years	84,390
			Hybrid		All	67,895
		Electric			All	42,814
	Medium	Carburator	3-Ways	PCV Tank	>8 years	66
		Single FI	3-Ways		>8 years	36
		Multi FI	3-Ways		>8 years	14,554
			Euro III		>8 years	12,905
			Euro IV		>8 years	42,313
			Euro V		>8 years	14,119
			Euro 6		>8 years	36,881
					4-8 years	32,793
					<4 years	66,776
			Hybrid		All	47,405
		Electric			All	32,969
Sml Engines	Light	4-Cycle	None	PCV	>8 years	32,042
			Euro III		>8 years	32,589
			Euro IV		>8 years	189,233
			Euro V		>8 years	113,307
			Euro 6		>8 years	237,737
					4-8 years	190,758
					<4 years	502,310
			Hybrid		All	19,308
		Electric			All	12,175
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	23,204
		Fuel-Injection	3-Ways		>8 years	262,238
			Euro III		>8 years	147,072
			Euro IV		>8 years	169,535
			Euro V		>8 years	47,459
			Euro 6		>8 years	65,520
					4-8 years	30,890
					<4 years	4,013
			Hybrid		All	7,855
		Electric			All	5,463

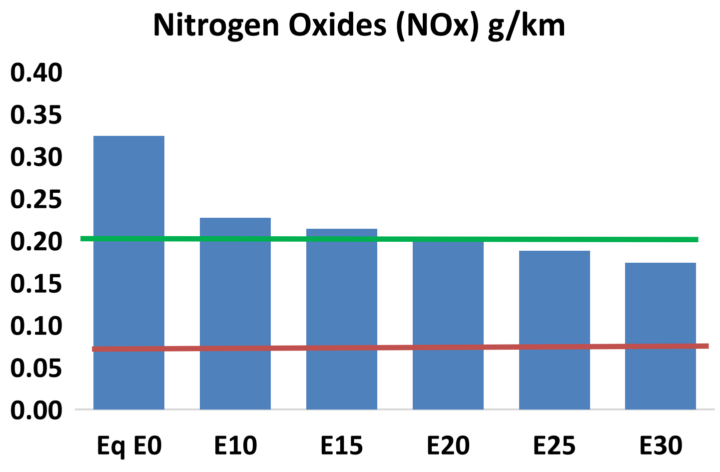
Vehicle Fleet: 3,207,228
Gasoline Vehicle Fleet: 1,641,194

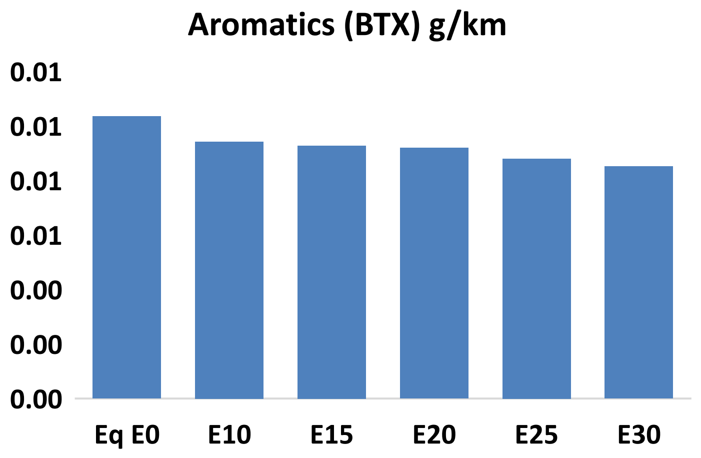
Source: Eurostat, ACEA

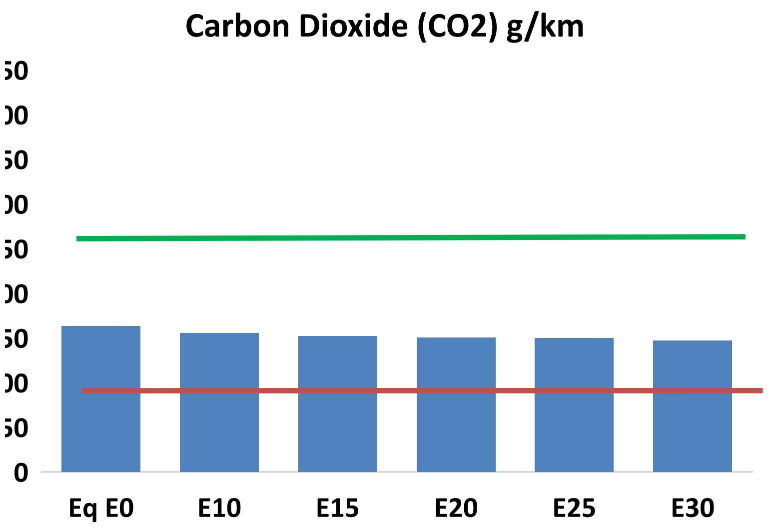
Ireland – Vehicular Emissions

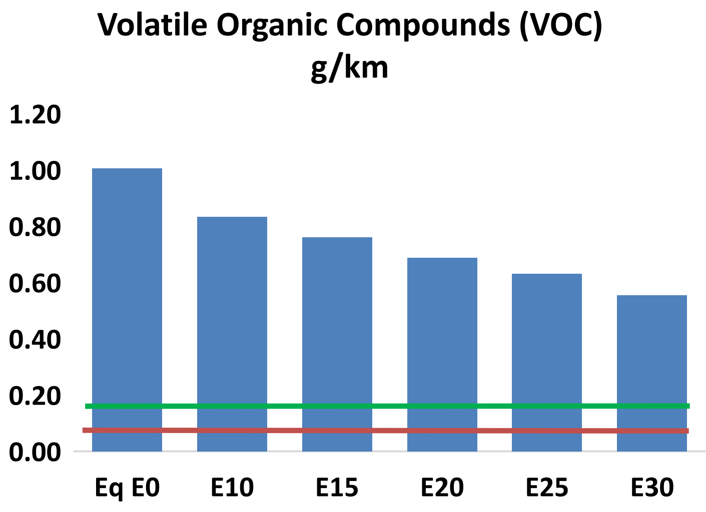
TIER III BIN 125 United States
Euro 6 Standard

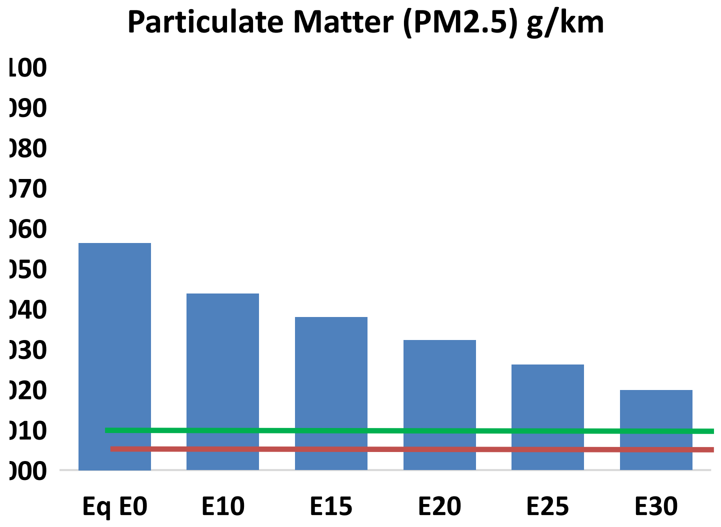












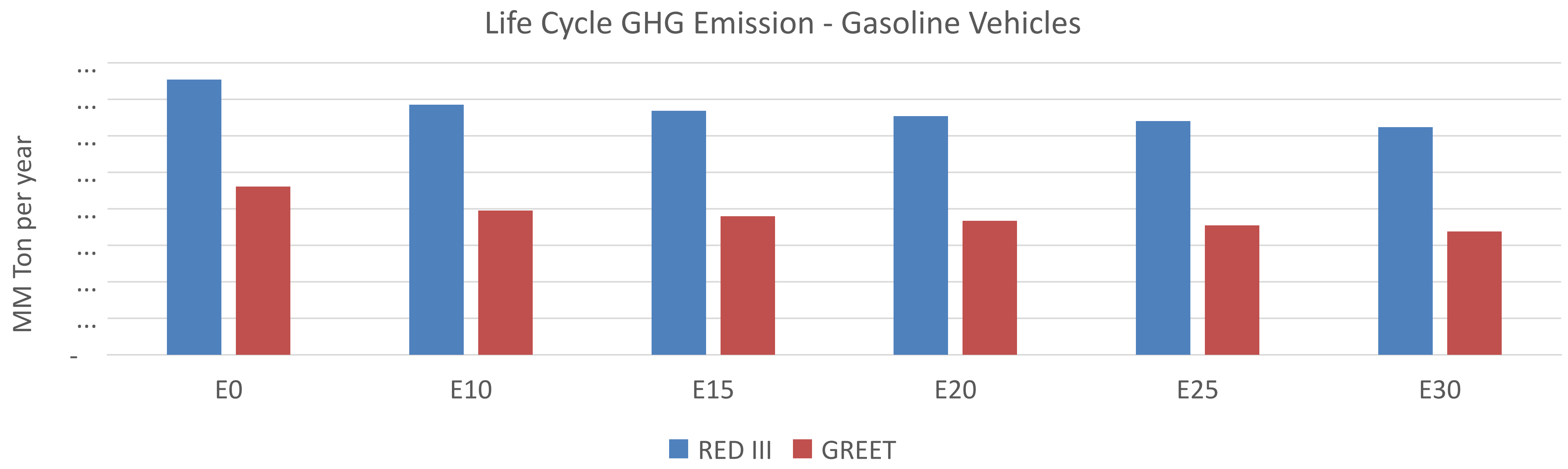
Source: Faro90

Ireland – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	211,328	188,438	165,548	145,494	124,716	108,559	92,049	-22%	-41%
CO2	6,629,738	6,459,259	6,298,251	6,171,623	6,108,963	6,075,101	5,963,120	-5%	-8%
VOC	40,883	37,376	33,869	30,924	27,928	25,614	22,561	-17%	-32%
NOx	13,148	9,861	9,204	8,675	8,181	7,637	7,061	-30%	-38%
SOx	74	68	63	58	53	49	43	-15%	-28%
NH3	2,749	2,722	2,695	2,708	2,738	2,760	2,777	-2%	0%
N2O	118	117	116	117	119	121	122	-1%	2%
CH4	8,968	8,968	8,968	9,103	9,300	9,471	9,632	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	216	203	190	180	170	162	152	-12%	-22%
Acetaldehyde	725	832	1,221	1,860	2,534	2,895	3,421	68%	249%
Formaldehyde	2,850	2,771	3,230	3,793	3,973	4,396	4,785	13%	39%
Benzene	420	404	383	377	374	357	346	-9%	-11%
PM2.5	447	397	347	301	256	208	157	-22%	-43%
PM10	1,837	1,631	1,425	1,238	1,051	853	646	-22%	-43%

Source: Faro90

Ireland – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	70.0
Target @ 2035 (MMT/year)	39.4
Delta	30.6

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	14.1%	17.5%	20.3%	23.1%	26.6%
RED II/III	0.0%	9.1%	11.4%	13.3%	15.1%	17.4%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	2.1%	2.6%	3.1%	3.5%	4.0%
RED II/III	0.0%	2.2%	2.8%	3.3%	3.7%	4.3%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90